

Dataset Report

State Farm Distracted Driver Detection Dataset

1.0 Dataset overview

The selected dataset for this project is the State Farm Distracted Driver Detection dataset. It contains driver-facing images labelled according to the activity being performed by the driver. This makes the dataset suitable for developing an image classification system that can distinguish between safe driving and selected forms of distracted driving.

For this project, the dataset will act as the main training and evaluation source for the software component. It provides a controlled starting point for testing the driver behaviour classification stage before the output is connected to hardware-supported decision and alert logic.

2.0 Dataset files and structure

The dataset information is organised using CSV files and image folders. The most important file for understanding the labelled data is `driver_imgs_list.csv`. It records the driver subject, the class label and the image filename for each labelled training image.

File	Purpose
<code>driver_imgs_list.csv</code>	Lists labelled training images, including subject ID, class label and image filename.
<code>sample_submission.csv</code>	Shows the expected prediction format for test images; it does not contain true labels.

3.0 Dataset size

Dataset item	Value
Labelled training image records	22,424
Test-image rows in submission file	79,726
Driver subjects in labelled training data	26
Driver behaviour classes	10

4.0 Behaviour classes

The dataset contains ten behaviour classes. One class represents safe driving, while the remaining classes represent different activities that may distract the driver.

Class	Driver behaviour	Images
c0	Safe driving	2,489
c1	Texting with right hand	2,267
c2	Talking on phone with right hand	2,317
c3	Texting with left hand	2,346
c4	Talking on phone with left hand	2,326
c5	Operating the radio	2,312
c6	Drinking	2,325
c7	Reaching behind	2,002
c8	Hair and makeup	1,911
c9	Talking to passenger	2,129

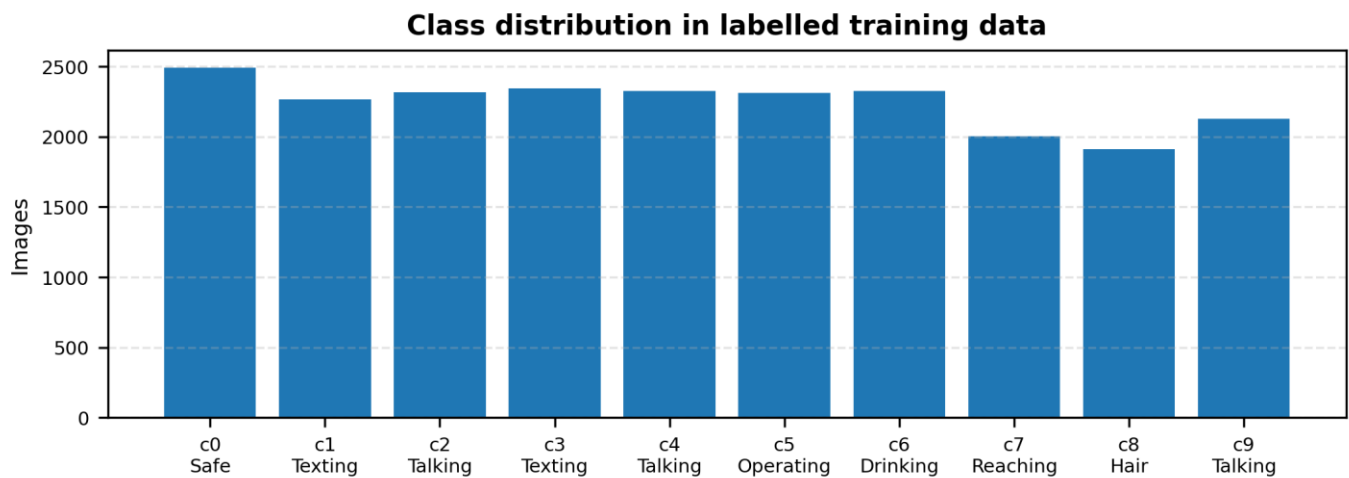


Figure 1: Class distribution in the labelled training data

5.0 Type of task and preparation approach

The dataset will be used as an image classification dataset. In this type of task, the system receives a full driver-facing image and predicts one behaviour class. The dataset does not provide bounding box labels for objects such as phones, hands or faces. For this reason, the project will treat the problem as classification rather than object detection.

The preparation stage will include checking file paths, mapping the class labels to their behaviour names, reviewing the class distribution and preparing the images for training and evaluation. Suitable preprocessing may include resizing, format conversion and normalisation. A driver-based split will be considered to reduce unfair testing caused by using images of the same driver in both training and evaluation sets.

6.0 Relevance to the project

The dataset directly supports the main classification requirement of the proposed FPGA-based driver distraction detection project. It provides labelled examples that can be used to train and test the behaviour recognition stage. The classification output can then be used by the prototype decision stage to identify whether the driver behaviour should be treated as safe or distracted.

The dataset will also support evaluation. Results such as class-level performance and a confusion matrix can show which behaviours are recognised reliably and which behaviours are more likely to be confused. These findings will help guide the discussion of the prototype and its limitations.

7.0 Limitations

The dataset is useful for controlled development, but it does not represent every real driving situation. Lighting conditions, camera position, vehicle interior and driver appearance may affect classification performance. The dataset also contains images rather than continuous real-time driving video, and it does not include object location labels.

These limitations will be acknowledged in the project report. The dataset will be used as a practical foundation for developing the classification component, while prototype testing can be used to discuss practical issues outside the original dataset environment.

8.0 Conclusion

The State Farm Distracted Driver Detection dataset is appropriate for this project because it provides labelled driver-facing images across ten driver behaviour classes. The available information includes 22,424 labelled training image records, 79,726 test-image rows in the submission-format file, 26 driver subjects and ten behaviour categories.

Overall, the dataset provides a clear basis for training and evaluating the image classification part of the project. The subject information and class distribution will be considered during preparation so that the results are interpreted carefully and fairly.

References

Kaggle (2016) State Farm Distracted Driver Detection. Available at: <https://www.kaggle.com/competitions/state-farm-distracted-driver-detection> (Accessed: 5 June 2026).